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Feedback

SQL PROJECT TO ANALYSE PIZZA SALES



ABOUT ME

I am passionate about learning SQL and applying it to real-world datasets. I enjoy exploring data, finding patterns, and presenting insights in a simple and meaningful way.

PROJECT OVERVIEW

This project focuses on analyzing pizza sales data through SQL queries. It highlights sales trends, top-performing pizzas, and customer buying behavior to support better decision-making and business growth.



DATA OVERVIEW

The dataset contains detailed information about pizza sales, including order IDs, pizza types, sizes, quantities, prices, and order dates. It provides both transactional and categorical data, allowing analysis of sales trends, customer preferences, and revenue patterns.





QUESTION FOR ANALYSIS

Basic:

- Retrieve the total number of orders placed.
- Calculate the total revenue generated from pizza sales.
- Identify the highest-priced pizza.
- Identify the most common pizza size ordered.
- List the top 5 most ordered pizza types along with their quantities.

Intermediate:

- Join the necessary tables to find the total quantity of each pizza category ordered.
- Determine the distribution of orders by hour of the day.
- Join relevant tables to find the category-wise distribution of pizzas.
- Group the orders by date and calculate the average number of pizzas ordered per day.
- Determine the top 3 most ordered pizza types based on revenue.

Advanced:

- Calculate the percentage contribution of each pizza type to total revenue.
- Analyze the cumulative revenue generated over time.
- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

ANALYSIS OUTPUT

```
-- Retrieve the total number of orders placed.
```

```
SELECT
```

```
    COUNT(order_id) AS total_no_of_orders
```

```
FROM
```

```
    pizza_db.orders;
```

total_revenue
817860.05

```
-- Calculate the total revenue generated from pizza sales.
```

```
SELECT
```

```
    ROUND(SUM(orders_details.quantity * pizzas1.price),  
          2) AS total_revenue
```

```
FROM
```

```
    orders_details
```

```
    JOIN
```

```
    pizzas1 ON orders_details.pizza_id = pizzas1.pizza_id;
```

total_no_of_orders
1350

ANALYSIS OUTPUT

```
-- Identify the highest-priced pizza.
```

```
SELECT
    pizza_types.name, pizzas1.price as highest_price_pizza
FROM
    pizza_types
    JOIN
    pizzas1 ON pizza_types.pizza_type_id = pizzas1.pizza_type_id
ORDER BY price DESC
LIMIT 1;
```

name	highest_price_pizza
The Greek Pizza	35.95

```
-- Identify the most common pizza size ordered.
```

```
SELECT
    pizzas1.size as most_common_pizza_type, COUNT(orders_details.quantity) as total_ordered
FROM
    pizzas1
    JOIN
    orders_details ON pizzas1.pizza_id = orders_details.pizza_id
GROUP BY most_common_pizza_type
ORDER BY total_ordered DESC
LIMIT 1;
```

most_common_pizza_type	total_ordered
L	18526

ANALYSIS OUTPUT

```
-- List the top 5 most ordered pizza types along with their quantities.

SELECT
    pizza_types.name as most_ordered_pizza, SUM(orders_details.quantity) AS total_quantities
FROM
    pizza_types
    JOIN
    pizzas1 ON pizza_types.pizza_type_id = pizzas1.pizza_type_id
    JOIN
    orders_details ON pizzas1.pizza_id = orders_details.pizza_id
GROUP BY pizza_types.name
ORDER BY total_quantities DESC
LIMIT 5;
```

most_ordered_pizza	total_quantities
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

ANALYSIS OUTPUT

```
-- Join the necessary tables to find the total quantity of each pizza category ordered

SELECT
    pizza_types.category as pizza_category, SUM(orders_details.quantity) AS total_quantities
FROM
    pizza_types
    JOIN
    pizzas1 ON pizza_types.pizza_type_id = pizzas1.pizza_type_id
    JOIN
    orders_details ON pizzas1.pizza_id = orders_details.pizza_id
GROUP BY pizza_types.category
ORDER BY total_quantities DESC;
```

pizza_category	total_quantities
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

ANALYSIS OUTPUT

```
-- Determine the distribution of orders by hour of the day.
```

```
SELECT
```

```
    HOUR(order_time) as hours, COUNT(order_id) AS total_distribution
```

```
FROM
```

```
    orders
```

```
GROUP BY HOUR(order_time)
```

```
ORDER BY total_distribution DESC;
```

hours	total_distribution
12	2520
13	2455
18	2399
17	2336
19	2009
16	1920
20	1642
14	1472
15	1468
11	1231
21	1198
22	663
23	28
10	8
9	1

```
-- Join relevant tables to find the category-wise distribution of pizzas.
```

```
select category, count(name) as distribution from pizza_types
```

```
group by category;
```

category	distribution
Chicken	6
Classic	8
Supreme	9
Veggie	9

ANALYSIS OUTPUT

```
-- Group the orders by date and calculate the average number of pizzas ordered per day
```

```
SELECT
```

```
    AVG(total_count) as avg_no_of_pizza_ordered_per_day
```

```
FROM
```

```
    (SELECT
```

```
        orders.order_date,
```

```
        SUM(orders_details.quantity) AS total_count
```

```
    FROM
```

```
        orders
```

```
    JOIN orders_details ON orders.order_id = orders_details.order_id
```

```
    GROUP BY orders.order_date) AS pizza_per_day;
```

avg_no_of_pizza_ordered_per_day

138.4749

ANALYSIS OUTPUT

```
-- Determine the top 3 most ordered pizza types based on revenue.
```

```
SELECT
```

```
    pizza_types.name as most_ordered_pizza,  
    SUM(pizzas1.price * orders_details.quantity) AS revenue
```

```
FROM
```

```
    pizzas1
```

```
        JOIN
```

```
    orders_details ON pizzas1.pizza_id = orders_details.pizza_id
```

```
        JOIN
```

```
    pizza_types ON pizza_types.pizza_type_id = pizzas1.pizza_type_id
```

```
GROUP BY pizza_types.name
```

```
ORDER BY revenue DESC
```

```
LIMIT 3;
```

most_ordered_pizza	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

ANALYSIS OUTPUT

```
-- Calculate the percentage contribution of each pizza type to total revenue.

SELECT
    pizza_types.category,
    ROUND(SUM(orders_details.quantity * pizzas1.price) / (SELECT
        SUM(orders_details.quantity * pizzas1.price) AS total
    FROM
        orders_details
        JOIN
        pizzas1 ON orders_details.pizza_id = pizzas1.pizza_id) * 100,
    2) AS percentage
FROM
    pizzas1
    JOIN
    orders_details ON orders_details.pizza_id = pizzas1.pizza_id
    JOIN
    pizza_types ON pizzas1.pizza_type_id = pizza_types.pizza_type_id
GROUP BY pizza_types.category
ORDER BY percentage DESC;
```

category	percentage
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

ANALYSIS OUTPUT

```
-- Analyze the cumulative revenue generated over time.

select order_date, round(sum(revenue) over (order by order_date),2) as cumm_revenue
from
(select orders.order_date,sum(orders_details.quantity * pizzas1.price) as revenue
from orders_details
join pizzas1 on orders_details.pizza_id = pizzas1.pizza_id
join orders on orders_details.order_id = orders.order_id
group by orders.order_date) as sales;
```

order_date	cumm_revenue
2015-01-01	2713.85
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.35
2015-01-11	25862.65

ANALYSIS OUTPUT

```
-- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

select category, name as top_3_pizza, revenue from
(select category, name, revenue, rank() over(partition by category order by revenue desc) as r from
(select pizza_types.category,pizza_types.name,sum(pizzas1.price * orders_details.quantity) as revenue
from orders_details join pizzas1 on orders_details.pizza_id = pizzas1.pizza_id
join pizza_types on pizzas1.pizza_type_id = pizza_types.pizza_type_id
group by pizza_types.category,pizza_types.name) as a) as b
where r<4;
```

category	top_3_pizza	revenue
Chicken	The Thai Chicken Pizza	43434.25
Chicken	The Barbecue Chicken Pizza	42768
Chicken	The California Chicken Pizza	41409.5
Classic	The Classic Deluxe Pizza	38180.5
Classic	The Hawaiian Pizza	32273.25
Classic	The Pepperoni Pizza	30161.75
Supreme	The Spicy Italian Pizza	34831.25
Supreme	The Italian Supreme Pizza	33476.75
Supreme	The Sicilian Pizza	30940.5
Veggie	The Four Cheese Pizza	32265.700000000065
Veggie	The Mexicana Pizza	26780.75
Veggie	The Five Cheese Pizza	26066.5

THANK YOU

Thank you for reviewing my
project

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